

PROJECT 1

Wazuh Windows Log Monitoring & Suspicious Login Detection

SOC Analyst L1 Practical Project Documentation

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Platform	Wazuh Open Source Security Platform
Server	Ubuntu Server
Monitored Endpoint	Windows-PC (Agent 004)
Endpoint IP	
Wazuh Version	v4.14.7
Investigation	Windows Event ID 4625 — failed logon

Purpose: document the lab setup, Windows endpoint onboarding, alert detection, and the SOC-style investigation of failed Windows logon events using Wazuh.

1. Project Overview

Objective: Build a small SOC monitoring lab in which an Ubuntu Server hosts Wazuh and a Windows endpoint sends security events to the Wazuh platform. The investigation focuses on detecting and analyzing Windows failed logon events (Event ID 4625).

What was demonstrated

- Wazuh Manager, Indexer and Dashboard services running on Ubuntu Server.
- Windows-PC successfully registered as Wazuh Agent 004.
- Endpoint telemetry visible in Wazuh with Windows 10 Pro and agent version v4.14.7.
- Threat Hunting used to filter authentication failures.
- Event ID 4625 details inspected at field level.
- A SOC-style verdict was formed using frequency, logon type, process context and source-address information.

Lab Architecture

Component	Role	Evidence
Ubuntu Server	Wazuh Manager + Indexer + Dashboard	Service status screenshots
Wazuh Agent 004	Collects Windows security telemetry	Agent list / agent details
Windows-PC	Monitored endpoint	IP – Windows 10 Pro
Threat Hunting	Searches and investigates alerts	Event 4625 query and details

2. Ubuntu Server & Wazuh Service Validation

The Ubuntu Server was used as the Wazuh server. The captured terminal evidence shows the three core services in the lab: Wazuh Manager, Wazuh Indexer and Wazuh Dashboard.

```
just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

15 updates can be applied immediately.
15 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

ayush@ubuntu-server:~$ sudo systemctl status wazuh-manager
[sudo] password for ayush:
● wazuh-manager.service - Wazuh manager
   Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-08-21 09:08:32 UTC; 19h ago
     Process: 919 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
    Tasks: 247 (limit: 9431)
   Memory: 1.1G (peak: 1.1G)
      CPU: 2min 51.786s
   CGroup: /system.slice/wazuh-manager.service
           └─1329 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             1330 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             1331 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             1334 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             1338 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             1373 /var/ossec/bin/wazuh-authd
             1389 /var/ossec/bin/wazuh-db
             1428 /var/ossec/bin/wazuh-execd
             1442 /var/ossec/bin/wazuh-analysisd
             1453 /var/ossec/bin/wazuh-syscheckd
             1467 /var/ossec/bin/wazuh-remoted
             1502 /var/ossec/bin/wazuh-logcollector
             1519 /var/ossec/bin/wazuh-monitord
             1531 /var/ossec/bin/wazuh-modulesd

Aug 21 09:08:26 ubuntu-server env[919]: Started wazuh-syscheckd...
Aug 21 09:08:28 ubuntu-server env[919]: Started wazuh-remoted...
Aug 21 09:08:28 ubuntu-server env[919]: Started wazuh-logcollector...
Aug 21 09:08:29 ubuntu-server env[919]: Started wazuh-monitord...
Aug 21 09:08:29 ubuntu-server env[1528]: 2026/08/21 09:08:29 wazuh-modulesd:router: INFO: Loaded router module.
Aug 21 09:08:29 ubuntu-server env[1528]: 2026/08/21 09:08:29 wazuh-modulesd:content_manager: INFO: Loaded content_manager module.
Aug 21 09:08:29 ubuntu-server env[1528]: 2026/08/21 09:08:29 wazuh-modulesd:inventory-harvester: INFO: Loaded Inventory harvester module.
Aug 21 09:08:30 ubuntu-server env[919]: Started wazuh-modulesd...
Aug 21 09:08:32 ubuntu-server env[919]: Completed.
Aug 21 09:08:32 ubuntu-server systemd[1]: Started wazuh-manager.service - Wazuh manager.
ayush@ubuntu-server:~$
```

Figure 1 — Wazuh Manager service: systemctl status shows the service as active (running) and enabled.

```
└─1329 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
  1330 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
  1331 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
  1334 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
  1338 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
  1373 /var/ossec/bin/wazuh-authd
  1389 /var/ossec/bin/wazuh-db
  1428 /var/ossec/bin/wazuh-execd
  1442 /var/ossec/bin/wazuh-analysisd
  1453 /var/ossec/bin/wazuh-syscheckd
  1467 /var/ossec/bin/wazuh-remoted
  1502 /var/ossec/bin/wazuh-logcollector
  1519 /var/ossec/bin/wazuh-monitord
  1531 /var/ossec/bin/wazuh-modulesd

Aug 21 09:08:26 ubuntu-server env[919]: Started wazuh-syscheckd...
Aug 21 09:08:28 ubuntu-server env[919]: Started wazuh-remoted...
Aug 21 09:08:28 ubuntu-server env[919]: Started wazuh-logcollector...
Aug 21 09:08:29 ubuntu-server env[919]: Started wazuh-monitord...
Aug 21 09:08:29 ubuntu-server env[1528]: 2026/08/21 09:08:29 wazuh-modulesd:router: INFO: Loaded router module.
Aug 21 09:08:29 ubuntu-server env[1528]: 2026/08/21 09:08:29 wazuh-modulesd:content_manager: INFO: Loaded content_manager module.
Aug 21 09:08:29 ubuntu-server env[1528]: 2026/08/21 09:08:29 wazuh-modulesd:inventory-harvester: INFO: Loaded Inventory harvester module.
Aug 21 09:08:30 ubuntu-server env[919]: Started wazuh-modulesd...
Aug 21 09:08:32 ubuntu-server env[919]: Completed.
Aug 21 09:08:32 ubuntu-server systemd[1]: Started wazuh-manager.service - Wazuh manager.
ayush@ubuntu-server:~$ sudo systemctl status wazuh-indexer
Unit wazuh-indexer.service could not be found.
ayush@ubuntu-server:~$ sudo systemctl status wazuh-indexer
● wazuh-indexer.service - wazuh-indexer
   Loaded: loaded (/usr/lib/systemd/system/wazuh-indexer.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-08-22 04:14:21 UTC; 5min ago
     Docs: https://documentation.wazuh.com
    Main PID: 918 (java)
      Tasks: 86 (limit: 9431)
   Memory: 1.6G (peak: 1.6G)
      CPU: 5min 10.911s
   CGroup: /system.slice/wazuh-indexer.service
           └─918 /usr/share/wazuh-indexer/jdk/bin/java -Xshare:auto -Dopensearch.networkaddress.cache.ttl=60 -Dopensearch.networkaddress.cache.negative.ttl=10

Aug 21 09:08:04 ubuntu-server systemd-entrypoint[918]: WARNING: System:setSecurityManager has been called by org.opensearch.bootstrap.OpenSearch (file:/usr/share/wazuh-indexer/lib/opensearch-2.11.0.jar)
Aug 21 09:08:04 ubuntu-server systemd-entrypoint[918]: WARNING: Please consider reporting this to the maintainers of org.opensearch.bootstrap.OpenSearch
Aug 21 09:08:09 ubuntu-server systemd-entrypoint[918]: Aug 21, 2026 9:08:09 AM sun.util.locale.provider.LocaleProviderAdapter <clinit>
Aug 21 09:08:09 ubuntu-server systemd-entrypoint[918]: WARNING: COMPAT locale provider will be removed in a future release
Aug 21 09:08:12 ubuntu-server systemd-entrypoint[918]: WARNING: A terminally deprecated method in java.lang.System has been called
Aug 21 09:08:12 ubuntu-server systemd-entrypoint[918]: WARNING: System:setSecurityManager has been called by org.opensearch.bootstrap.Security (file:/usr/share/wazuh-indexer/lib/opensearch-2.11.0.jar)
Aug 21 09:08:12 ubuntu-server systemd-entrypoint[918]: WARNING: Please consider reporting this to the maintainers of org.opensearch.bootstrap.Security
Aug 21 09:08:12 ubuntu-server systemd-entrypoint[918]: WARNING: System:setSecurityManager will be removed in a future release
Aug 22 04:14:21 ubuntu-server systemd[1]: Started wazuh-indexer.service - wazuh-indexer.
lines 1-21/21 (END)
```

Figure 2 — Wazuh Indexer service: after the initial service-name check, the correct service is shown as active (running).

3. Wazuh Dashboard Service & Web Interface

```
ayush@ubuntu-server:~$ sudo systemctl status wazuh-dashboard
● wazuh-dashboard.service - wazuh-dashboard
   Loaded: loaded (/etc/systemd/system/wazuh-dashboard.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-08-21 09:07:51 UTC; 19h ago
     Main PID: 768 (node)
       Tasks: 11 (limit: 9431)
    Memory: 293.7M (peak: 344.1M)
       CPU: 55.227s
    CGroup: /system.slice/wazuh-dashboard.service
            └─768 /usr/share/wazuh-dashboard/node/bin/node --no-warnings --max-http-header-size=65536 --unhandled-rejections=warn /usr/share/wazuh-dashboard/s
Aug 22 04:14:40 ubuntu-server opensearch-dashboards[768]: {"type":"log","timestamp":"2026-08-22T04:14:40Z","tags":["info","plugins","wazuh","initialize"],"pid":768,"message":"Starting Wazuh OpenSearch Dashboards plugin"}
Aug 22 04:14:40 ubuntu-server opensearch-dashboards[768]: {"type":"log","timestamp":"2026-08-22T04:14:40Z","tags":["info","plugins","wazuh","initialize"],"pid":768,"message":"Wazuh OpenSearch Dashboards plugin initialized successfully"}
Aug 22 04:14:42 ubuntu-server opensearch-dashboards[768]: {"type":"log","timestamp":"2026-08-22T04:14:42Z","tags":["info","plugins","wazuh","monitoring"],"pid":768,"message":"Wazuh OpenSearch Dashboards plugin monitoring started"}
Aug 22 04:14:42 ubuntu-server opensearch-dashboards[768]: {"type":"log","timestamp":"2026-08-22T04:14:42Z","tags":["info","plugins","wazuh","cron-scheduler"],"pid":768,"message":"Wazuh OpenSearch Dashboards plugin cron-scheduler started"}
Aug 22 04:14:43 ubuntu-server opensearch-dashboards[768]: {"type":"log","timestamp":"2026-08-22T04:14:43Z","tags":["info","plugins","wazuh","cron-scheduler"],"pid":768,"message":"Wazuh OpenSearch Dashboards plugin cron-scheduler started"}
Aug 22 04:14:44 ubuntu-server opensearch-dashboards[768]: {"type":"log","timestamp":"2026-08-22T04:14:44Z","tags":["info","http","server","OpenSearchDashboards"],"pid":768,"message":"Wazuh OpenSearch Dashboards plugin http server started"}
Aug 22 04:14:47 ubuntu-server opensearch-dashboards[768]: {"type":"log","timestamp":"2026-08-22T04:14:47Z","tags":["info","plugins","wazuh","monitoring"],"pid":768,"message":"Wazuh OpenSearch Dashboards plugin monitoring started"}
Aug 22 04:14:47 ubuntu-server opensearch-dashboards[768]: {"type":"log","timestamp":"2026-08-22T04:14:47Z","tags":["info","plugins","wazuh","monitoring"],"pid":768,"message":"Wazuh OpenSearch Dashboards plugin monitoring started"}
Aug 22 04:15:01 ubuntu-server opensearch-dashboards[768]: {"type":"log","timestamp":"2026-08-22T04:15:01Z","tags":["info","plugins","wazuh","monitoring"],"pid":768,"message":"Wazuh OpenSearch Dashboards plugin monitoring started"}
Aug 22 04:15:01 ubuntu-server opensearch-dashboards[768]: {"type":"log","timestamp":"2026-08-22T04:15:01Z","tags":["info","plugins","wazuh","monitoring"],"pid":768,"message":"Wazuh OpenSearch Dashboards plugin monitoring started"}
lines 1-20/20 (END)
```

Figure 3 — Wazuh Dashboard service: active (running), with OpenSearch Dashboard/Wazuh plugin initialization messages.

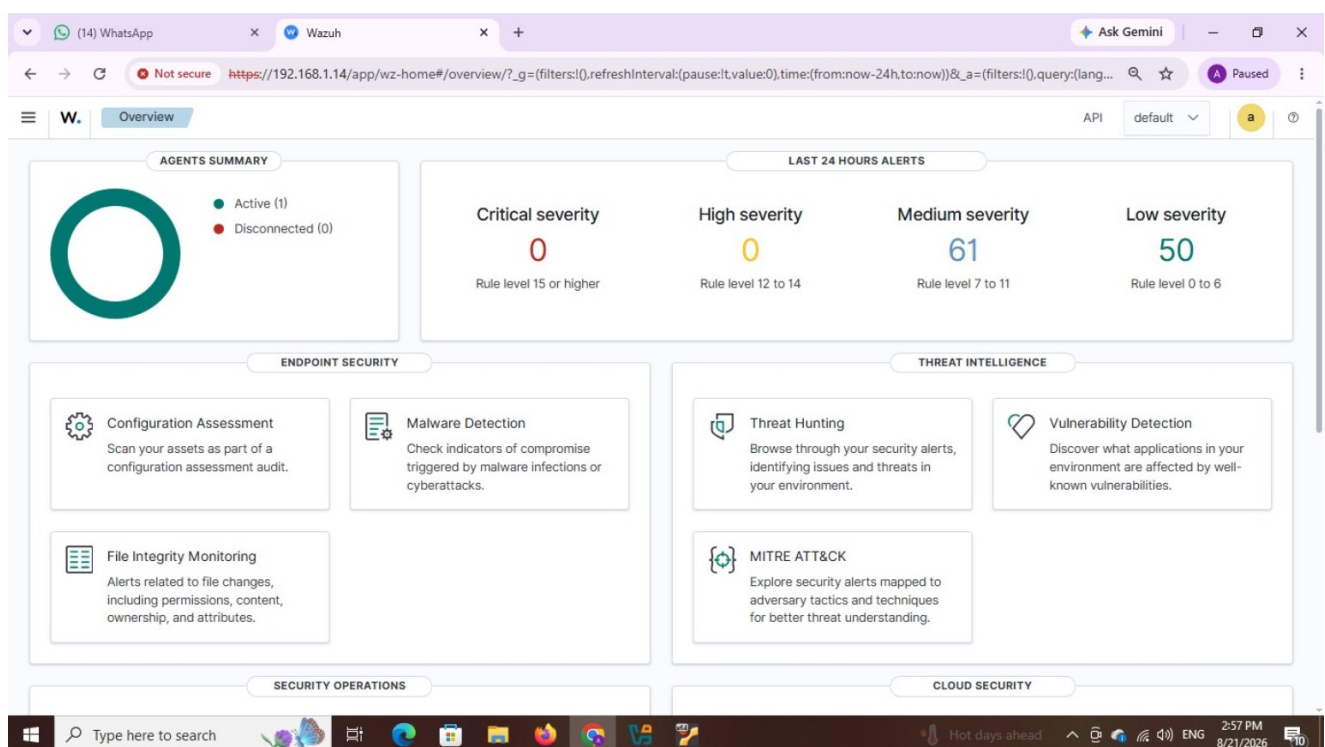


Figure 4 — Wazuh web dashboard: the platform is accessible and provides alert severity, endpoint security and threat-hunting modules.

Validation result: The service evidence and web dashboard together demonstrate that the Wazuh server stack was operational for the lab. The dashboard was then used for endpoint and alert investigation.

4. Windows Endpoint Registration

The Windows endpoint was onboarded to Wazuh as Agent 004. The endpoint screen identifies the device as Windows-PC, with IP address 192.168.1.4 and Wazuh version v4.14.7.

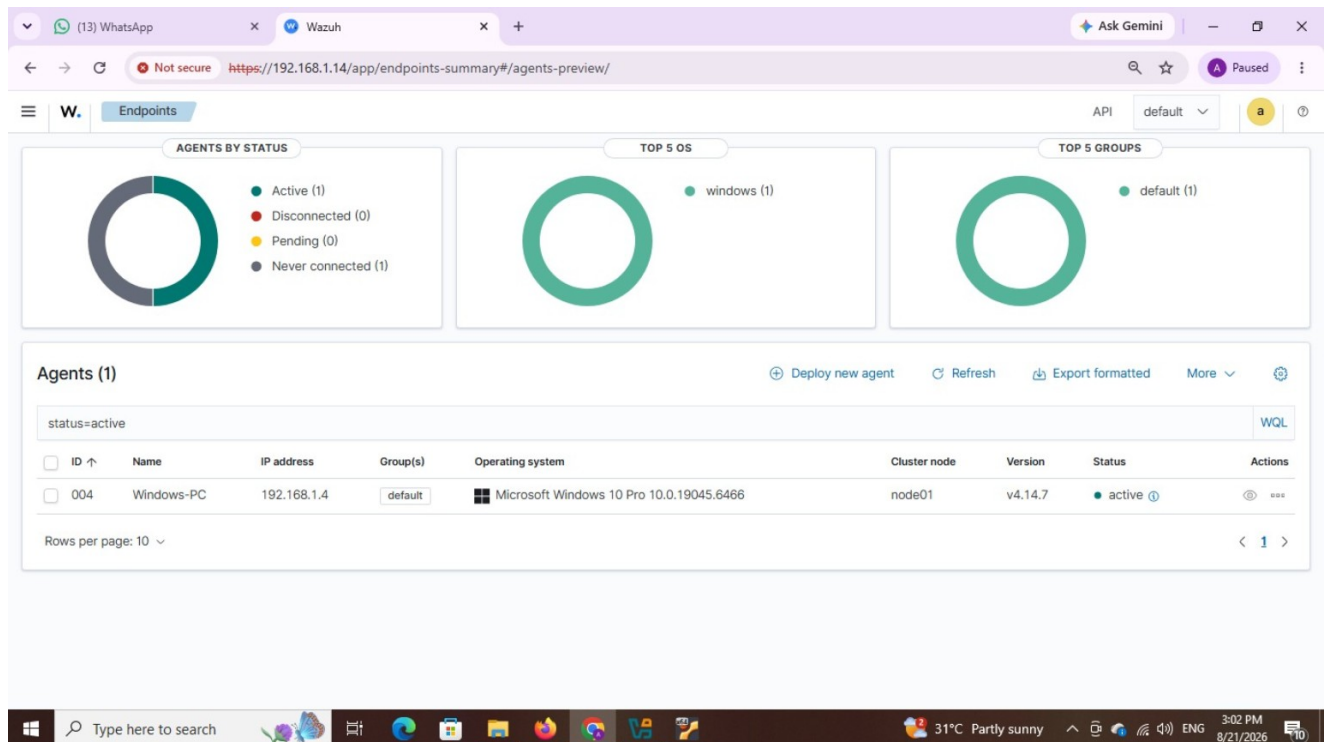


Figure 5 — Endpoints view: Agent 004 (Windows-PC) is listed as active with IP 192.168.1.4.

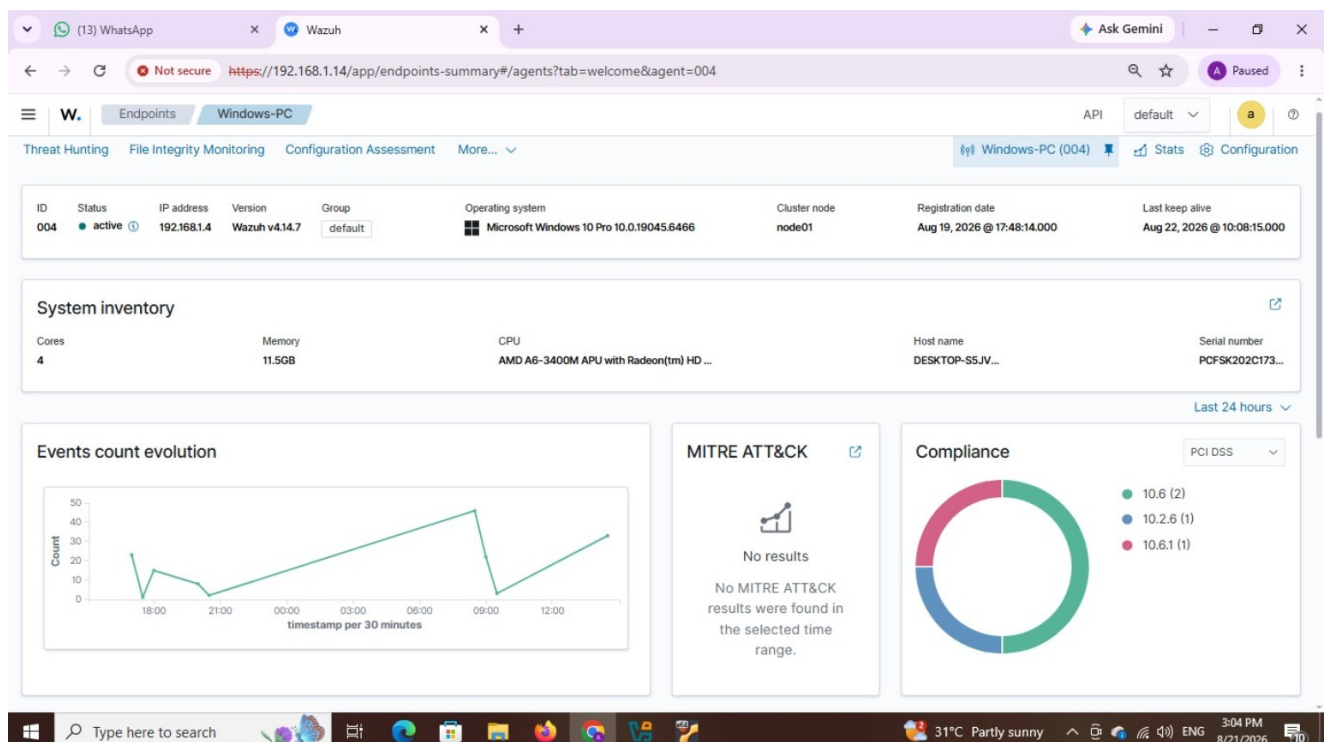


Figure 6 — Windows-PC agent details: active status, Windows 10 Pro, Wazuh v4.14.7, system inventory and event-count telemetry.

5. Threat Hunting & Alert Discovery

The Threat Hunting interface was used to inspect authentication activity for Agent 004. The dashboard screenshot shows authentication failure/success counters, while the event search was narrowed to the rule description “Logon Failure - Unknown user or bad password”.

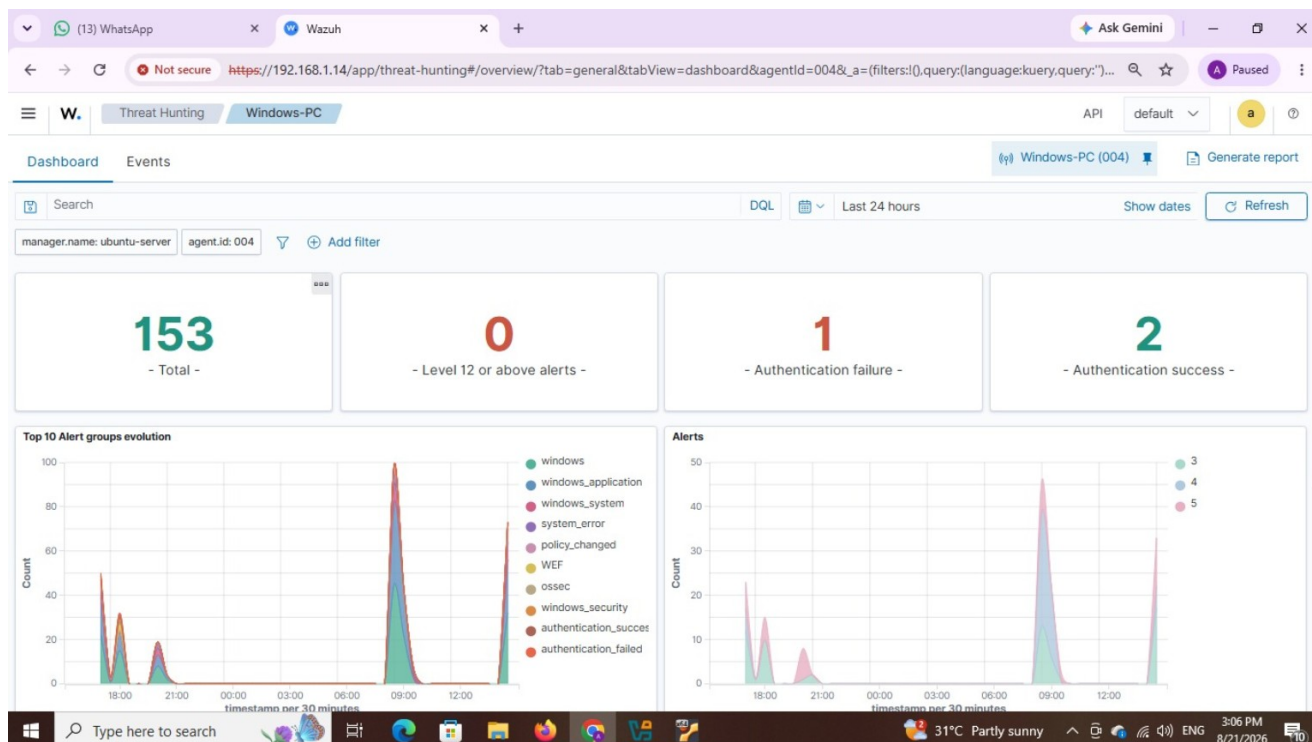


Figure 7 — Threat Hunting dashboard for Windows-PC (004): authentication activity is visible in the selected time window.

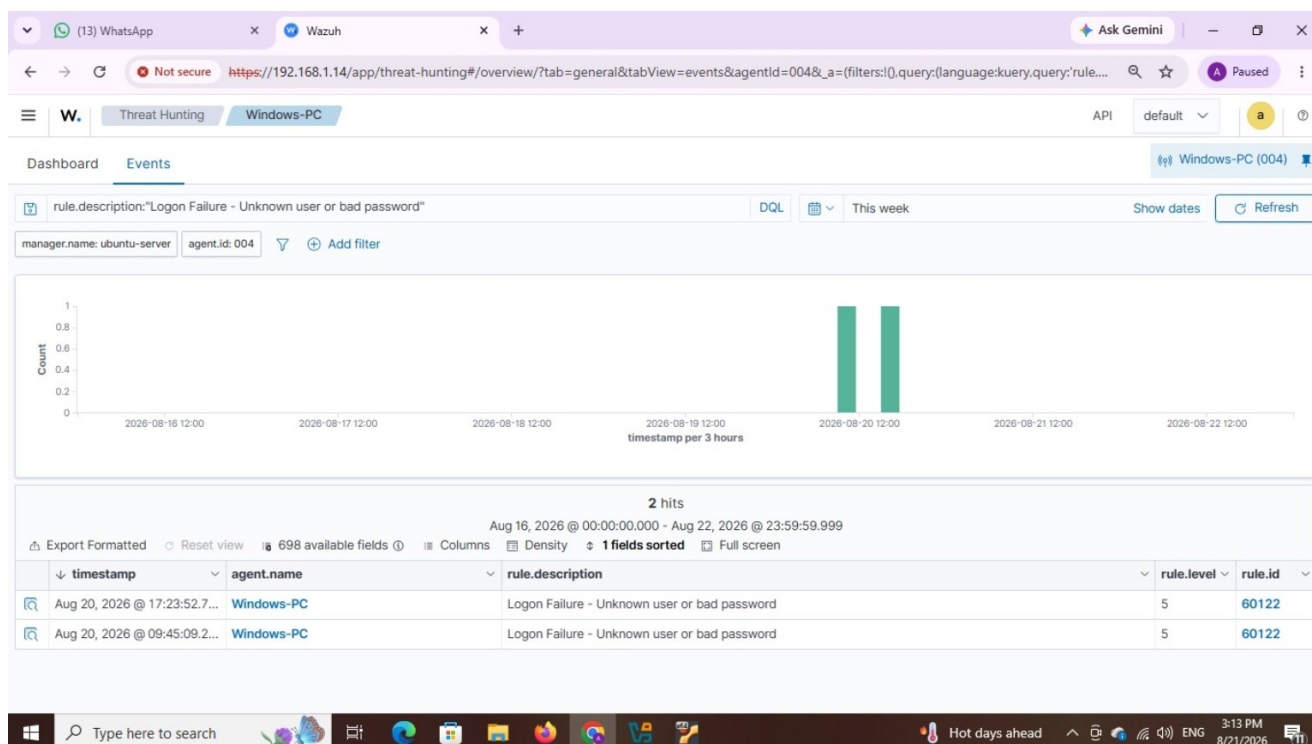


Figure 8 — Filtered Event view: two matching Event ID 60122/rule entries for “Logon Failure - Unknown user or bad password” are shown during the selected week.

Important time-window note: the dashboard and event-search screenshots use different time ranges (for example, Last 24 hours versus This week), so their counters should not be directly added together.

6. Event ID 4625 Investigation

Event ID 4625 is the Windows Security event associated with a failed login. The captured Wazuh document contains the Security channel, Event ID 4625, a failure reason, logon type and process context.

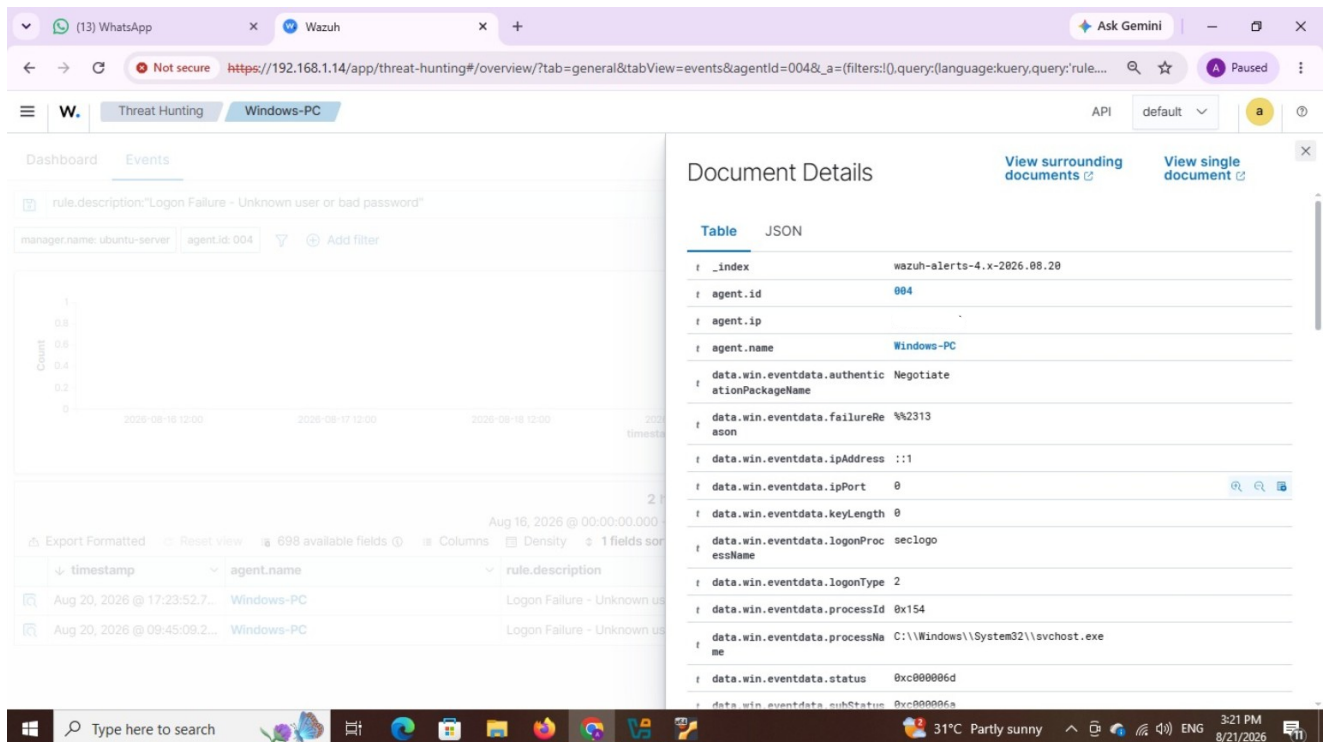


Figure 9 — Event 4625 field details: Agent 004, Windows-PC, failure reason %%2313, IP address ::1, logon process seclogon, Logon Type 2 and svchost.exe.

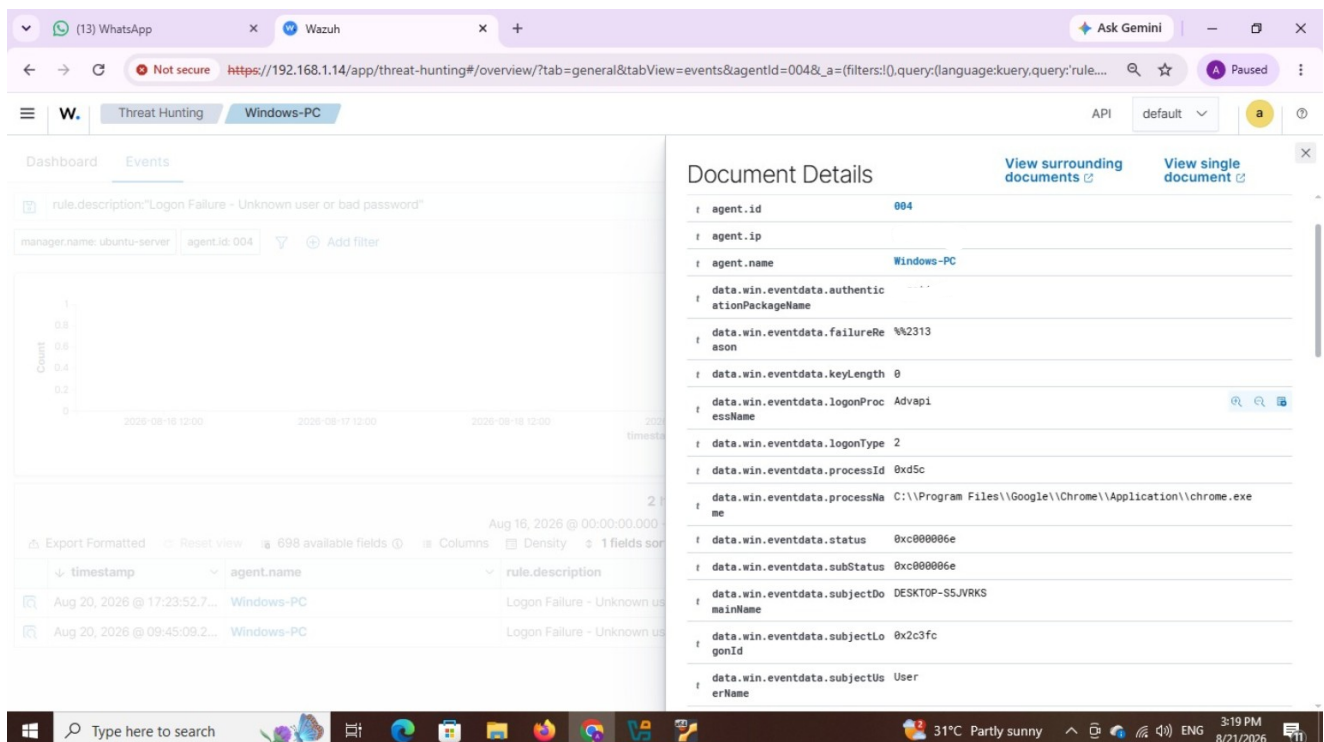


Figure 10 — Event 4625 process context from another captured record: Logon Type 2, Advapi, and Chrome.exe are shown.

7. Detailed SOC Analysis

Indicator	Observed evidence	SOC interpretation
Event ID	4625	Failed Windows logon; requires investigation.
Rule	60122	Wazuh classified the event as unknown user or bad password.
Frequency	2 matching events in the selected week	Low volume; not a clear brute-force pattern from the captured evidence.
Logon Type	2	Interactive/local logon context.
Source address	:::1 in the captured detailed record	Loopback address; does not indicate a remote source in that record.
Process context	svchost.exe in one record; Chrome.exe in another	Provides some process context; process alone is insufficient to prove malicious activity.
Severity	Wazuh rule level 5	Moderate informational signal; not a high/critical alert.

Verdict — Not confirmed malicious / low-confidence suspicious: The evidence confirms failed authentication attempts, but the captured records show Logon Type 2 and a loopback address in the detailed record, with only two matching events in the selected week. This is not enough evidence to label the activity as a confirmed attack. A SOC analyst should continue monitoring and correlate with the user's activity, Windows authentication history and any additional failed-logon bursts before escalation.

8. Recommended SOC Response

- Validate the account: confirm whether the affected username and failed logon were expected.
- Check for repetition: search for additional Event ID 4625 events from the same endpoint and account.
- Check for remote indicators: review future 4625 records for a non-loopback source address.
- Correlate process activity: compare the failed logon time with process creation and endpoint activity.
- Escalate only with supporting evidence: repeated failures, unusual accounts, remote sources, or suspicious process/network activity would increase confidence of malicious behavior.
- Document the disposition: retain the alert, evidence screenshot and analyst reasoning in the SOC ticket.

Key Learning Outcomes

- Understanding how a SIEM/XDR-style platform collects endpoint telemetry.
- Using Wazuh filters and rule descriptions to locate a security event.
- Reading Windows Event ID 4625 fields instead of judging an alert from the rule name alone.
- Separating an authentication failure from a confirmed security incident.
- Writing a concise SOC investigation with evidence, analysis and a defensible verdict.

This document is based on the screenshots captured during the lab. Where the screenshots do not establish a fact, the documentation intentionally avoids claiming it as confirmed.